

Derivatives: Various Rules and WRT

Numberbender | WORKSHEET



Name: _____ Date: _____ Score: _____

Find dy/dx for each function. Show all work.

Calculus I Worksheet 18

Find $\frac{dy}{dx}$ for #1 – 12.	
1. $y = \frac{2-x}{3x+1}$	2. $y = \ln \frac{e^x}{e^x - 1}$
3. $y = \frac{e^{\ln x}}{x}$	4. $y = \cos^2 x$
5. $y = x^{\ln x}$	6. $y = x(\ln^3 x)$
7. $x = t - \sin t$ and $y = 1 - \cos t$	8. $y = 3^{2x}$.
9. $y = x^2 \ln x$.	10. $y = \sqrt{5-2x}$
11. $y = \frac{x-3}{2-5x}$	12. $y = \sqrt{x^2 + 2x - 5}$
13. If $y = \sqrt{x^2 + 1}$, then find the derivative of y^2 with respect to x^2 .	14. If $y = x^2 + x$, then the derivative of y with respect to $\frac{1}{1-x}$ is ??
15. If $y = \sin x$, then find the derivative of y with respect to $\tan x$.	16. If $y = \sqrt{x^3 + 2x + 1}$, then find the derivative of y^2 with respect to $\cos x$.
17. If $f(x) = \ln x$, then find $f^{iv}(x)$. Note: $f^{iv}(x)$ is the 4 th derivative of $f(x)$.	18. If $f(x) = x^4 - 4x^2$, then $f^{(IV)}(2) =$
19. If $y = e^{-x^2}$, then $y''(0)$ equals ??	20. Write the equation of the line tangent to $y = 4x^2 + 3$ at $(2, 19)$



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Answer key — for instructor use only.

Answers:

1. $\frac{-7}{(3x+1)^2}$	2. $\frac{-1}{e^x - 1} = \frac{1}{1 - e^x}$	3. 0	4. $-2 \cos x \sin x = -\sin 2x$
5. $x^{\ln x} \bullet \frac{2 \ln x}{x}$	6. $x \ln^2 x + \ln^3 x$	7. $\frac{\sin t}{1 - \cos t}$	8. $2 \ln 3 \bullet 3^{2x}$
9. $2x \ln x + x = x(2 \ln x + 1)$	10. $\frac{-1}{\sqrt{5-2x}}$	11. $\frac{-13}{(2-5x)^2}$	12. $\frac{x+1}{\sqrt{x^2+2x-5}}$
13. 1	14. $(2x+1)(1-x^2)$	15. $\cos^3 x$	16. $\frac{-3x^2-2}{\sin x}$
17. $-6x^{-4}$	18. 24	19. -2	20. $y - 19 = 16(x - 2)$

